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Continuous Assessment Test (CAT) – I – July 2026

Programme	:	**B.Tech.**	Semester	:	**Fall Semester 2026-27**
Course Title	:	**Computer Architecture and Organization**	Course Code	:	**CSE2003**
Date/ Session	:	**10-8-2026 / S-I**	Slot	:	**A11+A12+A13**
Time	:	**1 ½ hours**	Max. Marks	:	**50**

Answer all the Questions

Q.No.	Sub. Sec.	Question Description	Mar
1		Explain the concept of Instruction Set Architecture (ISA). Apply your understanding to illustrate any three types of instruction set architectures with suitable assembly language examples	10
2		Evaluate the expression Y=A/B–(A–C*B) using **Stack Architecture** and **Accumulator Architecture**. Show the execution steps and compute the final value of **Y**. It is given A = 40 , B = 8 , C = 5.	10
3		Represent the following decimal numbers in both binary sign magnitude and twos complement using 16 bits: -128, -16, 40, -1; -92.	10
4		Analyse the execution of Booth's multiplication algorithm and the conventional binary multiplication method by multiplying 13 (multiplicand) and 11 (multiplier), where both numbers are represented using 6-bit binary numbers. Compare the two methods in terms of computational steps, handling of signed numbers, and operational efficiency	10
5		A processor supports **three instruction classes** with the following **Cycles Per Instruction (CPI)** values: Instruction Class / CPI: A – 3; B – 4; C – 6 Two different compilers generate the following instruction counts for the same program: Code Sequence in (million): Complier X – A 18, B 6, C 4; Complier Y – A 24, B 4, C 2 Assume that the processor operates at a **clock frequency of 800 MHz,** which code sequence will execute faster according to MIPS? And according to execution time?	1

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